

Test Project

Module B - HR Payroll and Leave Management System

Web Technologies

Contents

- Introduction
- Description of project and tasks
 - HR Admin Access
 - Employee Management
 - Public Holiday Management
 - Tax Bracket Configuration
 - Leave Requests
 - Payslip
 - JSON API Outputs
 - JSON API Example Outputs
 - Public-facing Pages
- Instructions to the Competitor
- Other

Introduction

We are going to create a payroll and leave management system for a company's HR department. The company runs a monthly payroll cycle and grants employees paid leave from an annual quota. When an employee takes leave on a day that is also a public holiday, the public holiday absorbs the day and it does not count against the employee's leave quota.

The payroll computation chains several inputs together: the number of effective working days in the pay month (weekdays minus public holidays falling on weekdays), the effective leave days consumed per employee in that same month, and a configurable progressive income tax schedule. The monthly tax deduction is derived from the employee's annual base salary applied across multiple tax brackets, then divided by twelve. All computed values are derived at generation time from live records. No pre-computed amounts are stored before a payslip is issued.

The system is operated by a single role, the HR admin, who manages employees, configures public holidays and tax brackets, administers leave requests, and generates monthly payslips. The general public can view the company's public holiday calendar without logging in.

The system uses two fixed leave types that are seeded into the database and cannot be managed through the UI: **Annual Leave** (12 days per year) and **Sick Leave** (12 days per year). All leave types are treated as paid; employees on approved leave are not penalised in their gross salary.

Description of project and tasks

In this project, we create an HR payroll and leave management system for the HR admin to manage employees, to process leave, and run monthly payroll.

It contains management web pages for employee records, public holiday configuration, tax bracket setup, leave request administration, and payslip generation. And a JSON API for querying employee leave balances.

This project should be reachable at http://wsXX.worldskills.org/XX_module_b/, where XX is the seat number.

HR Admin Access

The login page is at path `/XX_module_b/login`. The login page asks for an email and password to proceed with authentication. The email `admin@hrportal.com` and password `HRAdmin2026!` can be used to access the management system.

As a prototype product for 3 hours' work. We do not require robust authentication. The management system uses simple session-based checking.

If the submitted credentials are incorrect, the login page is shown again with the message Invalid email or password.

Accessing any management page without a valid session results in a 401 error.

After login, the admin is taken to the dashboard at `/XX_module_b/dashboard`. The dashboard shows the following counts, each as a link to the corresponding management page:

- Total number of active employees
- Number of pending leave requests
- Number of active employees who do not yet have any payslip for the current calendar month

Employee Management

HR admin can list, create, edit, and deactivate employees. The employee list is at `/XX_module_b/employees`. A particular employee is managed by accessing `/XX_module_b/employees/[employee_id]`, or by clicking the employee in the list. The Create employee form is at `/XX_module_b/employees/new`.

For each employee, the following data is stored and validated as follows:

Field	Rules
Full Name	Required.
Employee Code	Required. Must follow the format EMP0XX (e.g. EMP001, EMP002). Must be unique across all employees.
Email Address	Required. Must be a valid email address format. Must be unique across all employees.
Department	Required.
Base Monthly Salary	Required. Must be a positive integer.
Employment Start Date	Required. Must be a valid date.
Photo	Optional.

There could be one photo associated with each employee. The HR admin can upload and change this photo or remove the uploaded photo. The photo must be a JPEG, JPG, or PNG image with a file size not greater than 1 MB. When no photo is uploaded, there is a default placeholder image. The employee list page shows each employee's photo alongside their name and employee code.

An employee starts in active status. The HR admin can deactivate an employee. When an employee is deactivated:

- All **pending** leave requests for that employee are automatically **rejected**, with a note reading Employee deactivated.
- All **approved** leave requests where the start date is after today are automatically **cancelled**.
- Approved leave requests that have already started or are completed are not affected.

An employee cannot be deleted from the system. Active employees are shown in the main list. Inactive employees are shown in a separate section below, labelled Inactive Employees.

A deactivated employee cannot have new leave requests created for them and cannot have a payslip generated for them.

Public Holiday Management

HR admin can create, edit, and delete public holiday records. The Holiday List is at /XX_module_b/holidays, and the Create Holiday Form is at /XX_module_b/holidays/new. A particular holiday is edited by accessing /XX_module_b/holidays/[holiday_id].

For each public holiday, the following data is stored and validated as follows:

Field	Rules
Name	Required.
Date	Required. Must be a valid date. Each date may have at most one public holiday record - if the submitted date already exists, the form is rejected with the message A public holiday already exists for this date.
Description	Optional.

Public holidays are listed sorted by date ascending. The list shows all holidays for the current calendar year. Past holidays within the current year remain visible in the list.

Tax Bracket Configuration

HR admin can view and manage the progressive income tax bracket schedule at /XX_module_b/tax-brackets.

Each tax bracket stores and is validated as follows:

Field	Rules
Annual Income Lower Bound	Required. Must be a non-negative integer. The first bracket's lower bound must always be 0.
Annual Income Upper Bound	Must be a positive integer greater than the lower bound (exclusive).
Tax Rate	Required. Must be a decimal percentage with up to two decimal places (e.g. 15.00). Must be between 0.00 and 100.00.

After any add, edit, or delete operation, the system validates the full bracket set for continuity: no gaps and no overlaps are allowed. If validation fails, the change is rejected with the message Tax brackets must be gapless and non-overlapping. and no change is saved.

Leave Requests

The system uses two fixed leave types seeded into the database: **Annual Leave** (12 days per year) and **Sick Leave** (12 days per year). These are not managed through the UI. All leave types are treated as paid.

HR admin can view all leave requests at /XX_module_b/leave-requests. The list can be filtered by status and by employee. A new leave request is created at /XX_module_b/leave-requests/new, and a particular request is viewed at /XX_module_b/leave-requests/[request_id].

For each leave request, the following data is stored and validated as follows:

Field	Rules
Employee	Required. Must be an active employee. A deactivated employee cannot have new leave requests created for them.
Leave Type	Required. Must be one of the two seeded types: Annual Leave or Sick Leave.
Start Date	Required. Must be a valid date.
End Date	Required. Must be a valid date on or after the start date.
Reason	Optional.
Status	System-managed. Starts as pending on creation.
Rejection Note	Optional. Only applicable when status is rejected.

The HR admin creates a leave request on behalf of an employee by selecting the employee, leave type, date range, and optional reason. When submitted, the system computes the **effective days**: the number of weekdays (Mon–Fri) between the start and end dates inclusive that are not public holidays. This value is stored in the database immediately when the leave request is created. A

leave request cannot be submitted if the effective number of leave days **exceeds** the employee's remaining quota for that leave type in the current calendar year.

The remaining quota is: annual quota – (effective days of all pending requests + effective days of all approved requests) for the same employee, leave type, and calendar year. Because the effective days are stored on the record for both pending and approved requests, the quota check reads directly from the stored values.

Status transitions

A leave request starts in **pending** status. The allowed transitions are:

- pending → **approved** or **rejected**
- approved → **cancelled** (only if the start date is after today)
- rejected requests cannot be changed
- cancelled requests cannot be changed

When the HR admin **rejects** a pending request, they may optionally provide a rejection note. The rejection note is stored alongside the request and displayed on the request detail page when the status is rejected.

When a request is **approved**, the system **recomputes** the effective days using the same weekday-minus-public-holiday formula and **replaces** the stored value. This re-computation is necessary because public holidays may have been added or removed since the request was originally created, which would change the effective day count.

Schedule label

A schedule label is shown for every **approved** leave request wherever that request is displayed. It is computed on every page load and appears in both the leave request list table and the leave request detail page. No schedule label is shown for requests in any other status.

The schedule label rules are:

- Start date is in the future: Starts in [N] day(s)
- Today is within the leave period (start date ≤ today ≤ end date): Ongoing
- End date has passed (today > end date): Completed on [end date]

The end date in the Completed on label is displayed in the format Saturday, 15 March 2026.

Payslip

The payslip generation page is at `/XX_module_b/payslips/generate`. The HR admin selects a year and month, then generates payslips for all eligible active employees. An employee is eligible if they are active and do not already have a payslip for the selected period. Employees who already have one are skipped silently. Deactivated employees are never included.

Each generated payslip is immediately stored as issued. Once generated, a payslip cannot be edited, re-generated, or deleted.

Payslip computation

All values are computed at generation time from live records and stored as-is. The tax bracket configuration in effect at generation time is applied; it is not re-applied to already-generated payslips.

- **Working days in month:** count of weekdays (Mon–Fri) in the selected month, minus public holidays that fall on a weekday in that month.
- **Effective leave days:** for each approved leave request of the employee that overlaps with the pay month, count the weekdays within the overlapping range that are not public holidays. Sum across all overlapping requests.
- **Gross salary:** the employee's base monthly salary. Approved leave days do not reduce the gross salary.
- **Annual tax base:** gross salary $\times$ 12.
- **Annual tax liability:** apply progressive tax brackets to the annual tax base. For each bracket, multiply the portion of the annual tax base that falls within that bracket by the bracket's rate. Sum across all applicable brackets.
- **Monthly tax:** annual tax liability $\div$ 12.
- **Net salary:** gross salary – monthly tax.

Payslip listing

The payslips are listed at `/XX_module_b/payslips`. The list can be filtered by year and month, and by department. For each payslip row, the list shows: photo, full name, employee code, department, year, month (displayed as text, e.g. January, February), gross salary, monthly tax, net salary, and effective leave days. Each row links to the individual payslip detail page.

Payslip detail

A single payslip is viewed at `/XX_module_b/payslips/[payslip_id]`. The detail page shows: employee name, employee code, department, period (month and year), gross salary, monthly tax, net salary, working days in month, effective leave days, and the date the payslip was generated.

JSON API Outputs

The leave balance for an employee can be retrieved as JSON by accessing `GET /XX_module_b/api/employees/[employee_id]/leave-balance`. The response includes the employee's basic details and a list of leave types with the annual quota, days used, and remaining days for the current calendar year.

Days used for each leave type is the sum of effective days of all approved requests plus the effective days of all pending requests for that employee and leave type in the current year. Since effective days are stored on the record at creation (and updated on approval), this is a simple sum of the stored column across both statuses.

The endpoint returns a 404 status when the requested employee does not exist.

JSON API Example Outputs

Example leave balance output:

```
{
  "employee": {
    "id": 3,
    "name": "Budi Santoso",
    "code": "EMP001",
    "department": "Engineering",
    "status": "active"
  },
  "year": 2026,
  "leaveBalance": [
    {
      "leaveType": { "id": 1, "name": "Annual Leave" },
      "annualQuota": 12,
```

```
"used": 5,  
"remaining": 7  
,  
{  
  "leaveType": { "id": 2, "name": "Sick Leave" },  
  "annualQuota": 12,  
  "used": 0,  
  "remaining": 12  
}  
]  
}
```

Public-facing Pages

There is one public-facing page, accessible without authentication.

Company holiday calendar

The page at `/XX_module_b/public/holidays` shows all public holidays for the current calendar year, sorted by date ascending. The year filter is applied automatically on the server — no input or selection is needed from the visitor. Each row shows the date in the format Saturday, 15 March 2026, the holiday name, and the description if one is present. Past holidays are shown with a visual distinction from upcoming ones. The page title includes the current year, for example Public Holidays 2026.

Instructions to the Competitor

Database and model creation consideration

We would like to store the HR data in a database.

Please consider the relationships between employees, leave types, leave requests, public holidays, tax brackets, and payslips. An employee belongs to one department (stored as a string field on the employee). A leave request belongs to one employee and one leave type. A payslip belongs to one employee and covers one year-month period. A unique constraint on (employee_id, year, month) in the payslips table ensures that at most one payslip exists per employee per month.

Please provide the database dump as a file named db-dump.sql in the media folder. The dump must include table definitions with FK constraints and correct column types, and must be importable without modification.

Please provide the ER diagram as a file named erd.pdf in the media folder.

Please consider the normal form of the database.

Uploaded employee photos should be stored within the project directory at a consistent and predictable path. Do not store binary file content in the database.

Sample data is provided in media/seed.sql. Import it to populate the database before assessment.

Other

You may put your project in a sub-folder or different port. Please redirect to your destination from the path `wsXX.worldskills.org/XX_module_b/`.

The path mentioned in this document may be relative to your sub-folder.

Please also provide a file named `expert_readme.txt` to include the executing guide. You must provide this file even if you use the default executing path.

This project will be assessed using Firefox Developer Edition web browser.

Marking Summary

	Sub-Criteria	Marks
B1	HR Admin Access	2.25
B2	Database	0.50
B3	Employee Management	4.00
B4	Public Holiday Management	1.75
B5	Tax Bracket Configuration	2.50
B6	Leave Requests	4.25
B7	Payslip	5.75
B8	JSON API	1.25
B9	Public-facing Pages	0.75
	Total	23